

Hardware: input, output, storage and communication devices

★ **Golden**

One college period · 40–45 minutes · Overview of Computer System

What you will learn

- Classify devices as input, output, storage, processing or communication
- Describe keyboard, mouse, scanner, barcode, OCR, MICR, OMR, joystick, microphone, webcam
- Describe monitors, printers, plotter, speakers, projector
- Differentiate hardcopy and softcopy; impact and non-impact printers

Hardware. The physical, electronic and mechanical parts of a computer system.

Class	Job	Examples
Input	Send data into the computer	Keyboard, mouse, scanner, MICR, OMR, OCR, barcode, joystick, microphone, webcam, stylus, touch screen
Output	Send results to the user	Monitor, printer, plotter, speaker, projector, headphones
Storage	Keep data permanently or temporarily	HDD, SSD, USB, SD, CD/DVD/Blu-ray
Processing	Transform data	CPU, GPU
Communication	Send data between devices / networks	NIC, modem, switch, hub, router, gateway, fax

Input devices the board loves

- Barcode reader: optical scanner that reads product bars in a superstore for price and stock.
- OCR (Optical Character Recognition): reads printed text into editable text.
- OMR (Optical Mark Recognition): reads filled bubbles on MCQ sheets.
- MICR (Magnetic Ink Character Recognition): reads cheque numbers in banks.
- Scanner / data tablet / joystick / voice recognition / touch screen / webcam / microphone.

Output: hardcopy vs softcopy

	Hardcopy	Softcopy
Meaning	Printed permanent copy	Temporary display on screen
Device	Printer, plotter	Monitor, speaker, projector
Example	Marksheet on paper	Result on the college portal

Printers

Impact (hit the paper)	Non-impact (no hit)
Dot matrix, daisy wheel, line printer	Inkjet, laser, thermal
Noisy, carbon copies possible	Quiet, high quality
Cheap per page for multipart forms	Laser is fast for offices

WORKED EXAMPLE / BOARD STYLE

Plotter: an output device that draws large maps, CAD drawings and banners with pens. Board 2026 asked: "What devices are useful to print huge drawing or maps?" Answer: plotter (drum or flatbed).

HOW BIEK ASKS THIS

Ready differences: Hardcopy/Softcopy; Impact/Non-impact (four points); Barcode use in a superstore.

COMMON BOARD MISTAKE

A mouse is input, not output. A plotter is output, not a printer type you use for ordinary pages. Oracle is a DBMS, not an OS.

اردو میں سمجھیں

بارڈویئر چھونے والے حصے ہیں۔ ان پٹ کی بورڈ، ماؤس، بار کوڈ ریڈر، آؤٹ پٹ مانیٹر، پرنٹر، پلاٹر۔

بارڈ کاپی کاغذ پر چھپی کاپی ہے، سافٹ کاپی اسکرین پر۔ امپیکٹ پرنٹر کاغذ کو ٹکر مارتے ہیں، نان امپیکٹ (لیزر، انک جیٹ) نہیں مارتے۔

CHECK YOURSELF

1. Is OMR an input or output device?
2. Which printer makes carbon copies?

Answers: 1. Input — it reads marks from paper · 2. Impact / dot-matrix

MCQs

1. A scanning input device is:
 - a. Keyboard
 - b. Mouse
 - c. Track ball
 - d. OMR
2. Floppy disk and hard disk are types of:
 - a. Optical disk
 - b. Magnetic tape
 - c. Magnetic disk
 - d. Flash memory
3. A plotter is used to print:
 - a. Letters
 - b. Huge drawings and maps
 - c. Only photos
 - d. Sound

Key: 1=d, 2=c, 3=b

Short questions (3 marks each)

1. How can a barcode reader be useful in a superstore?
2. Differentiate hardcopy and softcopy.

3. Write any four differences between impact and non-impact printers.
4. What devices print huge drawings or maps?

Model short answers

1. It scans the bar code, fetches price and stock from the database, speeds billing and reduces typing errors.
2. Hardcopy is printed paper (permanent); softcopy is on screen or speaker (temporary, easy to edit).
3. Impact: strike paper, noisy, carbon copies, cheaper for forms. Non-impact: no strike, quiet, high quality, laser/inkjet.
4. Plotters (and large-format printers).

Long questions (board Section C)

1. Classify computer hardware. Explain input and output devices with examples.